Research Article

On the Internal Representation of Numerical Magnitude and Physical Size

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Abstract. A nascent idea in the numerical cognition literature – the analogical hypothesis (Pinel, Piazza, Bihan, & Dehaene, 2004) – assumes a common noisy code for the representation of symbolic (e.g., numerals) and nonsymbolic (e.g., numerosity, physical size, luminance) magnitudes. The present work subjected this assumption to various tests from the perspective of General Recognition Theory (GRT; Ashby & Townsend, 1986) – a multidimensional extension of Signal Detection Theory (Green & Swets, 1966). The GRT was applied to the dimensions of numerical magnitude and physical size with the following goals: (a) characterizing the internal representation of these dimensions in the psychological space, and (b) assessing various types of (in)dependence and separability governing the perception of these dimensions. The results revealed various violations of independence and separability with Stroop incongruent, but not with Stroop congruent stimuli. The outcome suggests that there are deep differences in architecture between Stroop congruent and incongruent stimuli that reach well beyond the semantic relationship involved.

Keywords: numerical cognition, dimensional interaction, General Recognition Theory, perceptual and decisional factors

When a numeral is presented for a brief inspection it provides the observer with rich information regarding a host of semantic and physical attributes such as magnitude, parity, or font size. A major challenge in numerical cognition is to determine how these dimensions are perceived and whether they interact in perception and cognition (Dehaene, 1997). Useful tools for addressing these questions have been selective attention paradigms such as the Stroop task (Stroop, 1935) and Garner's speeded classification task (Garner, 1974). Stroop and Garner effects have been often reported for numerical and physical dimensions (Henik & Tzelgov, 1982; Pansky & Algom, 1999, 2002; Schwarz & Heinze, 1998; Schwarz & Ischebeck, 2003). However, the internal representations that govern these interactions are still not well understood.

The present study proposes a novel framework for modeling and testing the internal representations of numbers. The dimensional relations between semantic magnitude and physical size are investigated here from the perspective of the General Recognition Theory (GRT; Ashby & Townsend, 1986). The GRT is a multidimensional generalization of Signal Detection Theory (SDT; Green & Swets, 1966; Macmillan & Creelman, 1991). Sustained by solid logical foundations and well worked out tools (Ashby, 1992; Fitousi, 2013; Kadlec & Townsend, 1992a, 1992b; Macho, 2007; Silbert & Thomas, 2013; Thomas, 2001a, 2001b; Townsend, Hu, & Evans, 1984; Townsend, Hu, & Kadlec, 1988), the GRT is capable of: (a) characterizing the internal representations of multidimensional stimuli, and (b) testing for various forms of dimensional interactions.

The Representation of Number

A number, say five, can be represented symbolically (i.e., "5") or nonsymbolically (i.e., a cardinal set of five dots). The former has been a relatively late cultural invention of literate societies, whereas the latter is presumably innate (Dehaene, 1997), requires no specific skill, and is shared by animals (Brannon & de Walle, 2001; Gallistel & Gelman, 1992; Nieder, Freedman, & Miller, 2002), infants (Wynn, 1992a, 1992b), and human beings from illiterate societies (Pica, Lemer, Izard, & Dehaene, 2004).

The *analogical hypothesis* proposes that symbolic numbers and nonsymbolic magnitudes (e.g., length, area, or luminosity) are processed in a similar fashion, with their formats being transformed automatically into noisy internal magnitude analogs (Dehaene, 1989, 1997; Dehaene & Akhavein, 1995; Dehaene, Dupoux, & Mehler, 1990; Moyer & Landauer, 1967; Piazza, Pinel, Bihan, & Dehaene, 2007; Shepard, Kilpatrick, & Cunningham, 1975). As Dehaene (1997) succinctly put it: "the brain forgets about the precise digits that led to this quantity."(p. 76).

Converging evidence in support of this idea comes from behavioral (Dehaene, 1989; Dehaene et al., 1990; Moyer &

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Landauer, 1967), developmental (Cordes, Gelman, Gallistel, & Whalen, 2001), imaging (Piazza et al., 2007; Pinel et al., 2004), and computational (Schwarz & Ischebeck, 2003; Verguts & Fias, 2004; Verguts, Fias, & Stevens, 2005) studies. For example, the time it takes to decide on the larger member of a pair of numbers is inversely related to the numerical distance separating them (i.e., the pair 2, 8 is responded to faster than the pair 6, 7). Similar distance effects were observed with nonsymbolic continua such as line length (Johnson, 1939), coin's size (Fitousi, 2010), animal's size (Pavio, 1975), and luminance (Pinel et al., 2004). The prevalence of the distance effect in both physical and symbolic dimensions has led researchers to believe that numerals are first transformed onto analog quantities, and are only then compared according to principles of what Moyer called "an internal psychophysics" or what Algom dubbed "memory psychophysics" (Algom, 1992; Dehaene et al., 1990; Link, 1990; Moyer, 1973; Restle, 1970; Shepard et al., 1975).

Neuroimaging studies (Fias, Lammertyn, Reynvoet, Dupont, & Orban, 2003; Pinel et al., 2004) have documented the activation of overlapping cortical sites for the two types of distance effects in the parietal cortex. Piazza, Pinel, Le Biahn, and Dehaene (2007) have shown, using an fMRI adaption technique, notation-independent coding of numerical quantity in the intraparietal sulcus. Observers recovered from adaptation to dot patterns or to Arabic numerals in a distance-dependent fashion whenever a new number was represented. All of this seems to support the idea that numerical magnitudes and nonsymbolic magnitudes are represented through the same noisy analog code.

Size Congruity Effects With Numerals

A central piece of evidence in favor of the analogical hypothesis comes from numerical Stroop-like tasks (Henik & Tzelgov, 1982). In this paradigm, observers are presented with pairs of numerals that can either be congruent or incongruent with respect to physical and numerical magnitude (e.g., 2–8 vs. 2–8). Observers are asked to select the physically larger member of the pair. It is often found that RTs in congruent conditions are faster than RTs in incongruent conditions. This *size congruity effect* (SCE) has been a richly documented phenomenon in the literature (Algom, Dekel, & Pansky, 1996; Besner & Coltheart, 1979; Fitousi, 2010; Fitousi & Algom, 2006; Foltz, Poltrock, & Potts, 1984; Girelli, Lucangeli, & Butterworth, 2000; Henik & Tzelgov, 1982; Schwarz & Heinze, 1998; Schwarz & Ischebeck, 2003; Takahashi & Green, 1983; Tzelgov, Meyer, & Henik, 1992; Vaid, 1985; Vaid & Corina, 1989). The interference of physical size with numerical magnitude has also been documented (Algom et al., 1996; Henik & Tzelgov, 1982; Pansky & Algom, 1999; Schwarz & Ischebeck, 2003).

There has been a general agreement among researchers that physical size and numerical magnitude make contact at some representational or processing level. However, the degree to which these dimensions interact as well as the locus/loci of interaction are still debated. A strong version of the analogical hypothesis assumes that the dimensions are translated first into an integrated internal analog which is then processed to activate a response (Piazza et al., 2007). According to this view, congruity effects with magnitude and size occur at an early processing stage. A softer version of the hypothesis proposes that the dimensions are processed in parallel, independent channels, and that both of them can elicit a separate response. On this account size and magnitude interactions are expected to be found in late (response) stages of processing (Algom et al., 1996; Tzelgov et al., 1992).

Evidence in support of either of these alternatives is rather mixed. For example, in an event-related potential study, Schwarz and Heinze (1998) concluded that: "numerical and size information are extracted in parallel, are both converted into an integrated representation... and potentially interact during these early parallel stages." (p. 1,167), whereas Pinel et al. (2004) argued that, "indeed, the finding of significant number/size interference in both behavior and fMRI provides positive evidence that the internal representations of number and size are not merely juxtaposed anatomically but share common neural resources" (p. 8).

Separability of Magnitude and Size

Algom and his colleagues (Algom et al., 1996; Fitousi, Shaki, & Algom, 2009; Pansky & Algom, 1999, 2002) have evaluated the (in)dependence of physical size and numerical magnitude from the perspective of Garner's speeded classification task (Garner, 1974). The Garner procedure is an operational test for assessing dimensional interaction between pairs of dimensions. The Garner task consists of three major experimental conditions. In the *baseline* condition, the relevant dimension (e.g., physical size) only varies from trial-to-trial, while the irrelevant dimension (e.g., numerical magnitude) is held constant. In the *correlated* condition values of the relevant dimension covary with values from the other dimension (e.g., small magnitude is combined with small physical size and vice versa). In the *filtering condition,* both dimensions vary from trial-to-trial in an orthogonal manner. The difference in RT between baseline and filtering is called *Garner Interference*. The difference in RT between a correlated block and baseline is dubbed *redundancy gain*. Pairs of dimensions that produce equality are considered to be integral dimensions because observers cannot process them separately. Pairs of dimensions that exhibit neither of these patterns are dubbed separable dimensions (Garner, 1974; Melara & Algom, 2003).

Do physical size and numerical magnitude combine to form integral or separable dimensions? According to Algom et al. (1996) the answer to this question depends on the relative discriminability of the dimensions. When numerical magnitude was made more discriminable than physical size the attributes appeared as integral dimensions. However, under condition of equal discriminability the

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attributes produced reliable redundancy gains but no Garner interferences. In view of these results, the dimensions cannot be considered as fully separable. This is because the classical definition of separable dimensions (Garner, 1974) requires the obtainment of neither redundancy gains *nor* Garner interference.

The theoretical insights drawn from the Garner and Stroop paradigms have informed research on numerical representation (Dehaene, 1997). However, these paradigms are operational. As such, they are susceptible to several known limitations (Ashby & Maddox, 1994; Maddox, 1992; Maddox & Ashby, 1996). First, it is not clear whether the separability or integrality concepts in these approaches apply at the level of representation or at the level of the processes that operate on them (Fitousi & Wenger, 2013; Maddox, 1992). Second, there is no principled way in these procedures of decoupling within-stimuli and across-stimuli dimensional interactions. Third, these paradigms do not take into account decisional processes that might affect perception. To achieve these goals, one needs to start with a theoretical model in which the psychological constructs and the possible internal representations are rigorously defined. The GRT (Ashby & Townsend, 1986) provides this model.

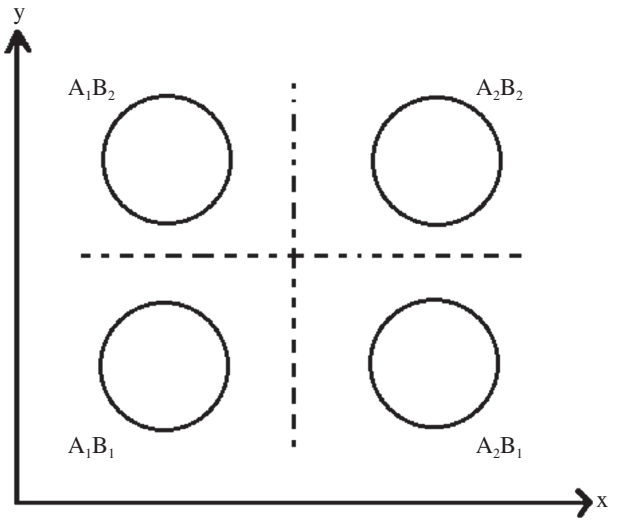


Figure 1. An example of the perceptual space in General Recognition Theory. Each of the equal-likelihood contours represents bivariate-Gaussian distributions. The broken lines represent the decision bounds. In this example perceptual independence, perceptual separability, and decisional separability hold.

General Recognition Theory

The GRT resulted from several decades of work on perceptual independence and separability (Fitousi, 2013; Garner & Morton, 1969; Tanner, 1956; Townsend et al., 1984, 1988). The theory affords the testing of hypotheses regarding the presence (or lack thereof) of interdependencies among dimensions or features of stimuli. Assume a classification task with four stimuli: A_1B_1, A_1B_2, A_2B_1, and A_2B_2. This set of stimuli is created by crossing two levels of two dimensions. A critical assumption made in GRT is that encoding of each stimulus engenders stochastic trial-by-trial perceptual effect (Ashby, 1992). The effect for a particular stimulus $A_i\ B_j$ is represented as a joint density function $f_{A_iB_j}(x, y)$. Bivariate-Gaussian distributions have been often assumed in GRT studies (Thomas, 1995). Hence, a 2×2 design with four stimuli leads to a perceptual space composed of four multidimensional distributions. When these distributions are cut horizontally at a fixed probability they can be represented by equal-likelihood contours. In addition, Observers assign a decisional bound with each perceptual region, and partition the perceptual space into four response regions. Figure 1 depicts such a GRT space.

Three psychological constructs are defined in GRT. Formal definitions can be found in many of the publications on GRT, including the seminal work by Ashby and Townsend (1986). Here I provide a brief verbal description of each of the constructs. *Perceptual Independence* (PI) refers to a form of within-stimulus statistical independence that holds if and only if the perceptual effects of components *A* and *B* are stochastically independent. When perceptual independence holds the equal-likelihood contours will appear as either circles or ellipses with the major and minor axes parallel to the coordinate axes in the perceptual space. When perceptual independence is violated, the ellipses will be tilted. *Perceptual Separability* (PS) holds if and only if the marginal perception of one dimension does not depend on the level of the other dimension. *Decisional Separability* (DS) holds if and only if the decision bound for one dimension remains invariant across levels of the other dimension.

The constructs of perceptual independence, perceptual separability, and decisional separability represent a decreasing strength of dimensional interaction. Violations of perceptual independence reflect the strongest form of interdependency among dimensions (Copeland & Wenger, 2006; Wenger & Ingvalson, 2002). This is because perceptual independence holds at the level of the individual stimulus, whereas violations of perceptual separability and decisional separability occur at the level of the stimuli ensemble. Of the last two constructs, it is decisional separability that represents the weakest form of dimensional interaction. Violations of decisional separability occur at a late decisional stage which is presumably amenable to influences of volitional factors (Maddox, 2001; Wenger & Ingvalson, 2002).

Modeling Number Perception With GRT

The modeling of the dimensional relations between magnitude and size requires specification of test stimuli and task. To create the stimuli, I crossed two levels of numerical

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magnitude (3, 7) with two levels of physical size (small, large). This factorial design results in four stimuli: 3, 3, 7, 7. On each trial the observer is presented with one of the four stimuli. His or her task is to report on the level of numerical magnitude and physical size of the presented stimulus. This complete identification task (Townsend et al., 1984) results in a confusion matrix in which the observer's errors are distributed across the off-diagonal cells. The matrice is then subjected to GRT analyses (Thomas, 2001a; Townsend et al., 1984, 1988).

Figure 2a–f depicts six possible GRT models that may underlie the internal representation of physical size and numerical magnitude. Again, in these panels cutting the bivariate distributions across the horizontal plane resulted in the equal-likelihood contours that inhabit the perceptual space. The major axis refers to perceptual effects of the numerical magnitude dimension (3, 7) assuming a left-to-right magnitude representation (Dehaene, 2007; Oeffelen & Vos, 1982; Verguts & Fias, 2004; Verguts et al., 2005).[1] The minor axis represents the perceptual effects of the dimension of physical size. Piecewise decision bounds appear as dashed lines.

Figure 2a describes a benchmark model in which perceptual independence, perceptual separability, and decisional separability all hold for both dimensions. The equal-likelihood contours appear as circles with their centers forming a rectangle. The decision bounds are parallel to the major and minor axes. In this model the representations of physical size and numerical magnitude are perceptually independent and perceptually and decisionally separable. In Figure 2b, the perceptual evidence for physical size is affected by the level of numerical magnitude, while the reverse does not hold true. Thus, perceptual separability is violated on the dimension of physical size, but not on the dimension of numerical magnitude. This state of affairs is graphically illustrated as a trapezoid created by the centers of the equal-likelihood contours.

Figure 2c and d shows the case of *mean shift integrality* (Maddox, 1992; Silbert & Thomas, 2013), in which perceptual separability is violated in a unique fashion. The means of the distributions on one dimension are shifted relative to each other, but the distances between each pair remain equal, as in the perceptually separable case. Two possible models of mean shift integraility are described in Figure 2c and d. In the former, perceptual separability is violated on numerical magnitude, whereas in the latter perceptual separability is violated on physical size.

In Figure 2e, the location of the decision bound for numerical value changes with the level of physical size. In contrast, the decision bound for physical size does not change as a function of numerical magnitude, indicating that decisional separability is violated for judgments of numerical magnitude, but not for judgments of physical size. In Figure 2f, the equal-likelihood contours are elliptic and tilted with their centers forming a rectangle. The decision bounds are parallel to the axes. This state of affairs implies that only perceptual independence is violated. This occurs within all four stimuli – congruent and incongruent alike. Note that physical size and numerical (i.e., semantic) magnitude combine to form Stroop-like stimuli. Font size can be either congruent (e.g., 3 vs. 7) or incongruent (e.g., 3 vs. 7) with semantic magnitude. Congruent stimuli in the GRT space are represented by the top-right and bottom-left equal-likelihood contours. Incongruent stimuli are depicted by the bottom-right and top-left equal-likelihood contours. The GRT allows one to test for each stimulus if perceptual independence is violated.

A strong version of the analogical hypothesis (Piazza et al., 2007) would predict violations of perceptual independence and perceptual separability; A soft version of the hypothesis (Algom et al., 1996; Pansky & Algom, 1999) would predict perceptual independence and perceptual separability, along with possible violations of decisional separability. The experiment was performed to decide between the two versions.

Before we proceed to the experiment, a methodological comment is in order. Recent work on GRT has shown that failure of decisional separability is not identifiable in Gaussian GRT models (Mack, Richler, Gauthier, & Palmeri, 2011; Silbert & Thomas, 2013). Silbert and Thomas (2013) have argued persuasively that GRT representations in which perceptual separability holds and decisional separability is violated cannot be mathematically (and empirically) distinguished from GRT representations in which perceptual separability is violated but decisional separability holds. These authors suggested that, "the assumption of decisional separability is both practically useful and less presumptive than assumptions of perceptual separability and independence" (p. 15). They further argued that, "... by making this assumption, we can engage powerful theoretical and statistical machinery to draw strong inferences about perceptual separability and independence" (p. 16).

In view of Silbert and Thomas' (2013) argument, the present work adopts the assumption of decisional separability. On the one hand, this assumption poses a certain limitation on the scope and type of GRT models that can be tested; on the other hand, it opens the door for a powerful investigation. Because the GRT constructs are logically independent (Ashby & Townsend, 1986), there is still room for rigorous modeling and testing of the role played by violations of perceptual independence and perceptual separability in the processing of physical size and numerical magnitude dimensions. This will allow me to distinguish between the strong and soft versions of the analogical hypothesis (Piazza et al., 2007). The two versions make clear predictions regarding the best-fitting GRT models for physical size and numerical magnitude. Strong versions of the analogical hypothesis predict violations of perceptual independence and/or perceptual separability, whereas soft versions of the analogical hypothesis predict no violations whatsoever.

[1] Compelling evidence for the representation of numbers on a left-to-right mental number line comes from the Spatial-Numerical-Association-of-Response-Codes (the SNARC effect; Ben-Nathan, Shaki, Salti, & Algom, 2009; Dehaene, Bossinin, & Giraux, 1993; Fitousi et al., 2009).

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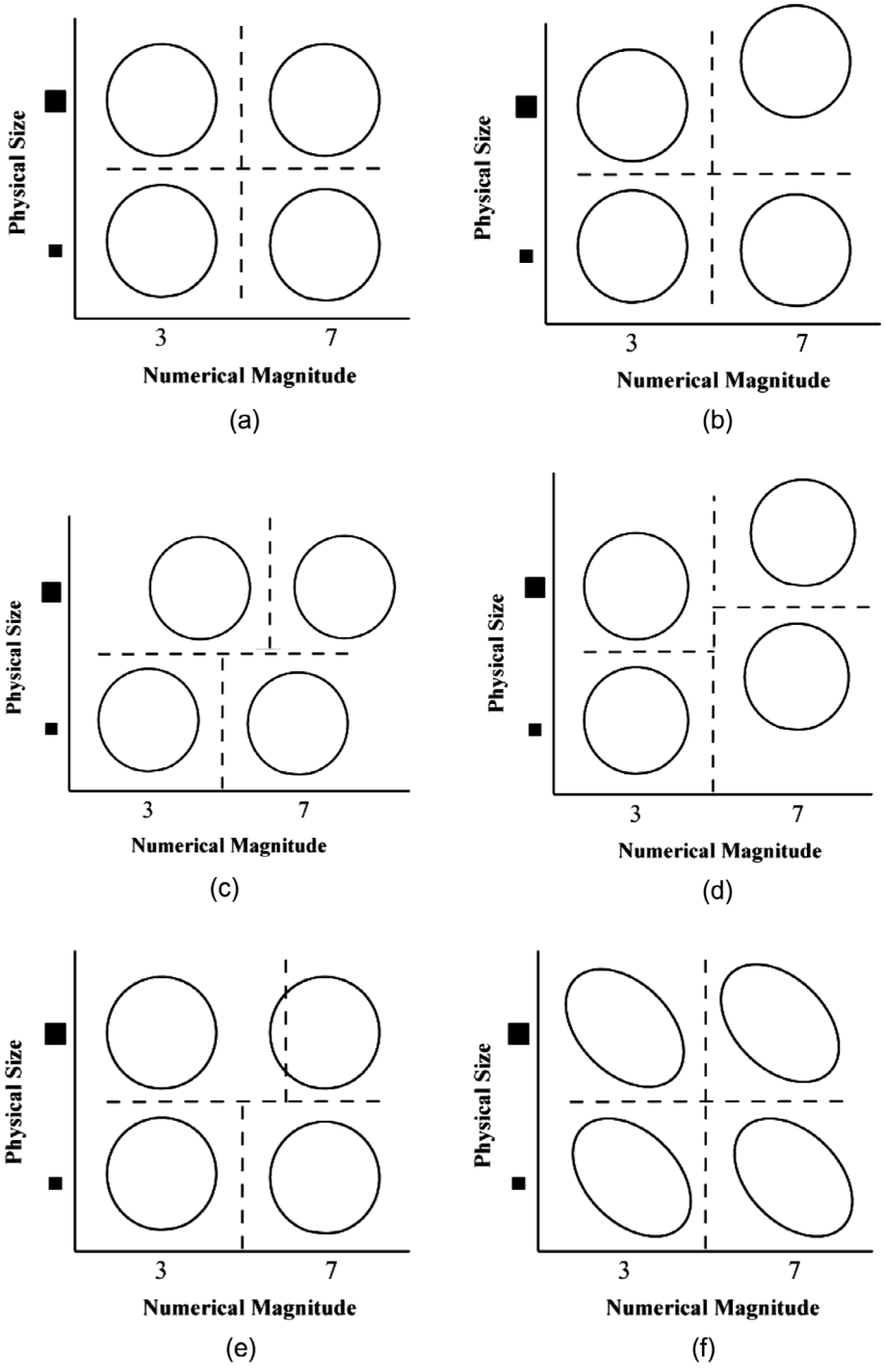


Figure 2. GRT models for magnitude and size. (a) Perceptual Independence, Perceptual Separability, and Decisional Separability hold, (b) Perceptual Independence and Decisional Separability hold, but Perceptual Separability is violated for judgments of physical size, (c) mean shift integrality-Perceptual Separability is violated for numerical magnitude but not for physical size, (d) mean shift integrality-Perceptual Separability is violated for physical size but not for numerical magnitude, (e) Perceptual Independence and Perceptual Separability hold, but Decisional Separability is violated for judgments of numerical magnitude, (f) Perceptual Separability and Decisional Separability hold, but Perceptual Independence is violated within all four stimuli.

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Method

Participants

Four volunteers from the Pennsylvania State University community were recruited for this Experiment. All observers had perfect or corrected to perfect vision. Each participant completed four consecutive days of testing.

Stimuli and Materials

The stimuli were the digits "3" and "7" (for observers 1 and 4) and the digits "2" and "8" (for observers 2 and 3). These numerals were presented in Times New Roman font and appeared in one of two possible font sizes, 0.4×0.9 cm (small) or 0.6×1.1 cm (large). Algom et al. (1996, see Experiment 1) reported on similar psychophysical values. These values combine to form stimuli in which physical size and numerical magnitude are of equal discriminability. Pilot work confirmed that with our values the dimensions are equally discriminable.

Stimuli were presented and responses were recorded using a PC-compatible computer with a 19 inch (diagonal) color CRT set to a resolution of $1{,}024 \times 768$ pixels. Stimulus onsets were synchronized with the vertical retrace of the monitor (refresh rate of 100 Hz), and responses were recorded using Macromedia's Authorware software (2007). The stimuli appeared gray over a black screen background on the center of the display. Observers viewed the display at a fixed distance of 76 cm through a black viewing tube, used to restrict the light available to observers' eyes to only the light emitted by the display. One end of the viewing tube was placed against the display while the other was placed against the observer's chin rest.

Design and Procedure

Observers were tested individually in a dimly lit room. The task required the identification of the levels of the numerical magnitude (2 or 8) and physical size (small, large). Each trial in each block began with a fixation point appearing at the center of the screen for 100 ms. This was replaced by a masking pattern for 50 ms, at which point the target number appeared for a duration that was set separately for each observer in each block, in order to obtain an overall accuracy level of between 65% and 75%. Another masking pattern appeared for 50 ms, after which the observers responded. The following response-assignment of keys was used: "z" for "3" (or "2") in small font, "x" for "3" (or "2") in large font, "n" for "7" (or "8") in small font, and "m" for "7" (or "8") in large font. No feedback was given.

Each experimental block consisted of 80 trials. Stimuli were presented with equal frequency in each block. Each experimental session consisted of 10 blocks. Prior to the experiment observers were presented with 10 training blocks (each consisting of 20 trials). On each trial one of the stimuli appeared on the screen for unlimited time, the stimuli was not masked. The observer had to press the correct button according to the key-assignment. The name of the correct button was given as a feedback after the observer responded. Observers had to score at least 95% correct responses to continue to the experimental blocks.

Results

The confusion-identification matrices for each observer appear in Table 1. I have used two statistical approaches to testing hypotheses regarding GRT constructs which have become routine in GRT studies (Cornes, Donnelly, Godwin, & Wenger, 2011; Fitousi & Wenger, 2013; Thomas, 2001a, 2001b; Wenger & Ingvalson, 2002, 2003). The first consists of a series of nonparametric tests, originally proposed by Ashby and Townsend (1986), in combination with a set of signal detection measures of sensitivity and bias (Kadlec & Townsend, 1992a, 1992b). The second approach involves the fitting of a set of parametric models to the identification/confusion data, using a hierarchical inference approach (Ashby & Lee, 1991; Thomas, 2001a). In combination, the two approaches allow for rigorous hypotheses testing of GRT constructs.

As noted earlier, recent work on GRT (Mack et al., 2011; Silbert & Thomas, 2013) has shown that in Gaussian GRT models decisional separability is not identifiable. In accordance with Silbert and Thomas' (2013) recommendation, decisional separability has been assumed in all subsequent analyses.

It is important to emphasize that the current work tested individual observers rather than the aggregate. Various researchers have documented considerable differences between the group and the individual results (Ashby, Maddox, & Lee, 1994; Estes, 1956; Fitousi & Wenger, 2011; Maddox, 1999), thereby making the testing of individual observers preferable to the aggregate.

Nonparametric Tests

The sampling independence (SI) test (Ashby & Townsend, 1986; Townsend, Hu, & Ashby, 1981; Townsend & Snodgrass, 1977; Townsend et al., 1984, 1988) confirms perceptual independence and decisional separability. Sampling independence is not a theoretical construct, but rather a property of the data that can be estimated directly from the confusion matrix and is tested at the level of the individual stimulus. Sampling independence in stimulus A_iB_j holds if and only if:

$$P(R_{a_2b_2}|A_iB_j) = \left[P(R_{a_2b_1}|A_iB_j) + P(R_{a_2b_2}|A_iB_j)\right] \times \left[P(R_{a_1b_2}|A_iB_j) + P(R_{a_2b_2}|A_iB_j)\right], \quad (1)$$

where $R_{a_ib_j}$ denotes that response a_ib_j was given. The conditional probabilities for other stimuli are defined in a similar fashion. Failures of sampling independence imply violations

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Table 1. Observed (top) and predicted (bottom) identification confusions

	Obs. 1				Obs. 2			
Stimulus	1	2	3	4	1	2	3	4
1(1, 1)	197	162	18	15	191	164	30	9
	197	162	18	14	200	156	21	16
2(1, 2)	9	353	10	20	29	332	20	16
	9	353	10	20	29	330	21	17
3(2, 1)	14	19	242	129	17	21	305	63
	14	19	242	129	16	18	309	64
4(2, 2)	1	11	53	327	8	18	71	296
	2	10	52	327	6	25	72	288
	Obs. 3				Obs. 4			
Stimulus	1	2	3	4	1	2	3	4
1(1, 1)	215	136	32	5	245	91	51	10
	223	128	23	13	250	85	45	15
2(1, 2)	71	328	20	16	136	185	51	7
	71	328	20	16	135	184	51	7
3(2, 1)	25	23	328	49	63	33	312	23
	25	23	328	49	59	30	318	23
4(2, 2)	9	13	94	236	48	27	140	178
	6	16	96	233	39	42	149	162

Note. The predicted values resulted, from the SI, MRI, macro-signal detection analyses, and the hierarchical model fitting procedure; Obs. = Observer.

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of perceptual independence and/or decisional separability (see Ashby & Townsend, 1986, Theorem 1). According to this logic, if decisional separability holds then failure of sampling independence implicates perceptual interdependency. The sampling independence test was modeled after Thomas (2001a, 2001b) and is constructed as a chi-square test (for a tutorial see Thomas, 2001b).

Table 2 gives the sampling independence test results per stimulus and observer. For all four observers, sampling independence dramatically fails mainly in the incongruent stimuli A_1B_2 and A_2B_1. This is indicated by the large χ^2 value. When sampling independence fails and decisional separability holds, one can safely conclude that perceptual independence is violated. The model-fitting procedure performed later on provides additional support to this conjecture.

Perceptual separability can be assessed with the Marginal Response Invariance (MRI) test (Ashby & Townsend, 1986; Silbert & Thomas, 2013). Marginal response invariance in an identification-confusion matrix holds when the probability of correctly recognizing one component does not depend on the physical level of the other. That is for $i = 1, 2$,

$$P(R_{a_ib_1}|A_iB_1) + P(R_{a_ib_2}|A_iB_1) = P(R_{a_ib_1}|A_iB_2) + P(R_{a_ib_2}|A_iB_2) \quad (2)$$

and for $j = 1, 2$,

$$P(R_{a_1b_j}|A_1B_j) + P(R_{a_2b_j}|A_1B_j) = P(R_{a_1b_j}|A_2B_j) + P(R_{a_2b_j}|A_2B_j). \quad (3)$$

Ashby and Townsend (1986) have shown that if perceptual separability and decisional separability hold for both components A and B, marginal response invariance holds. To test for marginal response invariance, the test statistics z was carefully constructed (cf. Silbert & Thomas, 2013; Thomas, 2001a, 2001b). Table 3 presents the results of the marginal response invariance tests. For most observers, the test failed for physical size, but held for numerical magnitude. The reader can verify this result by assessing the number of significant z statistics on each of the dimensions. These results suggest that perceptual separability is violated for physical size but not for numerical magnitude.

Table 2. Obtained χ^2 values for sampling independence tests

	Obs. 1	Obs. 2	Obs. 3	Obs. 4
Stimulus				
A_1B_1	0.001	$7.62^{\dagger}$	$9.212^{\dagger}$	3.11
A_1B_2	$57.15^{\ddagger}$	$68.33^{\ddagger}$	$28.46^{\ddagger}$	$40.80^{\ddagger}$
A_2B_1	$6.78^{\ddagger}$	$30.53^{\ddagger}$	$36.89^{\ddagger}$	$49.90^{\ddagger}$
A_2B_2	0.30	1.97	1.53	$9.70^{\dagger}$

Notes. H_0 = Sampling independence holds; A = Numerical magnitude; B = Physical size. This form of SI test was developed by Thomas (2001b). $^{*}p < .05$. $^{\dagger}p < .01$. $^{\ddagger}p < .001$.

Multidimensional Signal Detection Tests

The next series of tests for assessing perceptual separability (and decisional separability when it is not assumed) assume

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Table 3. Marginal response invariance tests

Condition	Obs. 1	Obs. 2	Obs. 3	Obs. 4
	b_1	b_1	b_1	b_1
Numerical size				
A_1B_1	0.915	0.901	0.904	0.846
A_1B_2	0.923	0.909	0.917	0.847
z	−0.394	−0.398	−0.634	−0.024
p Value	0.346	0.345	0.262	0.490
A_2B_1	0.082	0.093	0.112	0.222
A_2B_2	0.030	0.066	0.062	0.190
z	3.119	1.428	2.444	1.127
p Value	0.009†	0.076	0.007†	0.129
	a_1	a_1	a_1	a_1
Physical size				
A_1B_1	0.548	0.560	0.636	0.745
A_2B_1	0.633	0.793	0.830	0.870
z	−2.444	−7.031	−6.283	−4.564
p Value	0.007†	0.000‡	0.000‡	0.000‡
A_1B_2	0.048	0.123	0.209	0.493
A_2B_2	0.137	0.201	0.292	0.478
z	−4.301	−2.959	−2.699	0.417
p Value	0.000‡	0.001†	0.003†	0.338

Note. A = Numerical Magnitude; B = Physical Size. *$p < .05$. †$p < .01$. ‡$p < .001$.

normality and are based on signal detection measures. The pertinent measures are computed directly from the confusion-identification matrix in a similar fashion to computations of discriminablity (d') and response bias (c) in unidimensional SDT (Green & Swets, 1966; Macmillan & Creelman, 2005). The critical measures are differences in discriminability and response bias between pairs of marginal distributions. Statical tests were held using the methods developed by Thomas (2001a, see Appendix A, pp. 648–649), and Gourevitch and Galanter (1967) and are based on a construction of a z statistic.

Table 4 summarizes the results of the multidimensional-signal detection analyses. As can be noted, there are more significant differences in discriminabilities and response biases measures in judgments of physical size than judgments of numerical magnitude. These results do not necessarily implicate violations of perceptual separability (and decisional separability when it is not assumed) on either of the dimensions (see for, Kadlec & Townsend, 1992a, 1992b). One needs first to integrate the evidence from the marginal response invariance test and the model fits before making any inference. In addition, the assumption of decisional separability, adopted throughout, should be taken into consideration.

Silbert and Thomas (2013) have proposed a novel way of looking at the relations between marginal response invariance and the signal detection macro-analyses. They have argued that the two tests are essentially redundant, and therefore should lead to exactly the same inferences. If marginal response invariance holds for both levels of a dimension, then equalities of discriminabilities and response biases are implicated. Any observed discrepancy between the two tests should be attributed to statistical estimation errors. Silbert and Thomas (2013) showed that the marginal response invariance test is preferable, because it tends to produce lower levels of false alarms (i.e., inappropriately statistically significant tests). In the current analyses marginal response invariance failed for the two levels of physical size in the data of Observer 1, Observer 2, and Observer 3. The comparable multidimensional signal detection analyses for these Observers revealed inequalities of both d's and c's. Taken together, these outcomes suggest that perceptual separability is violated on physical size but not on numerical magnitude.

In conventional applications of GRT to empirical data (Cornes et al., 2011; Fitousi & Wenger, 2013; Richler, Gauthier, Wenger, & Palmeri, 2008; Wenger & Ingvalson, 2002, 2003) marginal response invariance and the signal detection macro-analyses are treated as two separate sources of information. Their implication relations are often summarized in a truth-table (see Table 5; see also Macmillan & Creelman, 2005, p. 262). The table is based on theoretical work by Ashby and Townsend (1986), and Kadlec and Townsend (1992a, 1992b). For example, if the marginal response invariance holds (i.e., True), but the marginal d'*'s* and c's are not equal (i.e., False, False), then perceptual separability and decisional separability are violated. Note that few inferences in this table do not lead to firm conclusions, a fact that has been noted in the original work on GRT (Ashby & Townsend, 1986), as well as by other researchers (Macho, 2007; Thomas, 1995).

Table 4. Multidimensional signal detection analysis

	Obs. 1		Obs. 2		Obs. 3		Obs. 4	
Parameter	d'	c	d'	c	d'	c	d'	c
Numerical magnitude								
AB_1	1.95	0.00	1.84	−0.01	1.78	0.03	1.26	0.09
AB_2	2.33	−0.15	2.00	−0.05	2.06	−0.05	1.34	0.05
p	0.00‡	0.00‡	0.07	0.29	0.00‡	0.06	0.48	0.50
Physical size								
A_1B	1.25	−0.54	0.92	−0.35	0.81	−0.16	0.47	0.22
A_2B	1.01	−0.26	1.17	0.00	1.06	0.14	0.83	0.37
p	0.03*	0.00‡	0.04*	0.00‡	0.04*	0.00‡	0.00‡	0.02*

Note. A = Numerical Magnitude; B = Physical Size; Obs. = Observer. *$p < .05$. †$p < .01$. ‡$p < .001$.

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Table 5. Logic relating marginal measures of performance to inferences regarding perceptual separability (PS) and decisional separability (DS)

	Evidence		Inference	
MRI	Equal marginal d'	Equal marginal c	PS	DS
T	T	T	T	T
T	T	F	T	F
T	F	T	F	T
T	F	F	F	F
F	T	T	T	?
F	T	F	T	F
F	F	T	F	?
F	F	F	F	?

Notes. Under evidence, T = equality in question held; F = equality in question did not hold. Under inference, T = no violation; F = violation. A question mark indicates uncertain inference.

This limitation can be addressed satisfactorily if the modeler uses the hierarchical model fitting procedure (Cornes et al., 2011; Fitousi & Wenger, 2013; Thomas, 2001a, 2001b). Moreover, the assumption of decisional separability made here strengthens the power of this combined approach (cf., Silbert & Thomas, 2013).

Table 6 summarizes inferences regarding the sampling independence, marginal response invariance, and multidimensional-signal detection tests in terms of Truth (i.e., pass) and False (i.e., failed) values. Table 7 gives the inferences regarding perceptual independence and perceptual separability. Virtually all inferences involving perceptual separability are false, pointing to violations of the construct, mostly in judgments of physical size. Because decisional separability has been assumed throughout, sampling independence becomes relevant to testing perceptual independence (see also, Kadlec & Townsend, 1992a, 1992b; Thomas, 2001a, 2001b; Wenger & Ingvalson, 2002, 2003). The fact that sampling independence failed in all of the incongruent stimuli points to violations of perceptual independence in these stimuli. The model fitting procedure performed next supports this conclusion.

Hierarchical Modeling

The model-fitting approach employed here was first proposed by Ashby and Lee (1991), and further developed by Thomas (2001a). The procedure is performed by defining a hierarchy of possible models, beginning with a null model in which perceptual independence, perceptual separability, and decisional separability hold. Each model in the hierarchy is nested within a constrained model posited higher in the branch. The hierarchy is then extended by systematically increasing the number of free parameters.

The simplest model thus contains four free parameters with the following mean vectors and covariance matrices:

$$\mu_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix},\ \mu_2 = \begin{bmatrix} 0 \\ \mu_{2,2} \end{bmatrix},\ \mu_3 = \begin{bmatrix} \mu_{3,1} \\ 0 \end{bmatrix},\ \mu_4 = \begin{bmatrix} \mu_{3,1} \\ \mu_{2,2} \end{bmatrix},$$

where the $\mu_{i,j}$ s are parameters to be estimated. The variance/covariance matrices are:

$$\sum\nolimits_i = \begin{bmatrix} \sigma_A^2 & \rho\sigma_A\sigma_B \\ \rho\sigma_A\sigma_B & \sigma_B^2 \end{bmatrix},$$

where i is the relevant stimulus and ρ is the correlation coefficient for each stimulus. In the basic model where perceptual independence holds the correlation coefficient is equal to zero ($\rho = 0$). In contrast, in models where perceptual independence is violated the correlation coefficient is either greater ($\rho > 0$) or smaller ($\rho < 0$) than zero. In addition, two piecewise decision bounds are specified as two parameters to be estimated, γ_1 and γ_2. Thus, the basic model contains four parameters.

I fit the models to the empirical identification confusion matrices separately for each observer using methods developed by Macho (2007). To make the number of GRT parameters smaller than the data points in the matrice, I have assumed equal variances and fixed the location of one of the bivariate distributions (A_1B_1) at (0, 0). These conventions have been adopted routinely in previous studies (see also Ashby & Lee, 1991; Thomas, 2001a, 2001b). To measure the model's fit to identification data, I used the maximum-likelihood function (see Thomas, 2001a, equation B2, p. 649). Decisions about the best-fitting model were based on the (—) log-likelihood for the model along with measures of fit, as explained next.

General models can show better absolute fit to the data than do restricted models. This can occur either because the general model has more free parameters, and therefore larger complexity, or because of its greater ability to capture real aspects of the data. To decide between these possibilities, one can use the *likelihood ratio test* (Thomas, 2001a).

Table 6. Summary of the marginal analyses and the nonparametric test of sampling independence (SI), for each of the four observers (Obs.)

	Obs. 1				Obs. 2				Obs. 3				Obs. 4			
Comparision	SI	MRI	d'	c	SI	MRI	d'	c	SI	MRI	d'	c	SI	MRI	d'	c
Mag × Size	2/4	F	F	F	3/4	T	T	T	3/4	F	F	T	3/4	T	T	T
Size × Mag	2/4	F	F	F	3/4	F	F	F	3/4	F	F	F	3/4	F	F	F

Notes. MRI = test of marginal response invariance; d' = test of marginal sensitivity; c = test of marginal response bias; T = equality holds; F = equality does not hold; the number in the SI column indicates the total number of failures stimuli for which SI failed (out of four possible). Mag = Numerical Magnitude; Size = Physical Size.

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Table 7. Summary of the inferences regarding Perceptual Independence (PI), Perceptual Separability (PS), and Decisional Separability (DS) for the dimensions of Numerical Magnitude and Physical Size for each of the four observers (Obs.)

	Obs. 1			Obs. 2			Obs. 3			Obs. 4		
Comparision	PI	PS	DS	PI	PS	DS	PI	PS	DS	PI	PS	DS
Mag × Size	F	F	T	F	T	T	F	F	T	F	T	T
Size × Mag	F	F	T	F	F	T	F	F	T	F	F	T

Note. T = the pertinent construct is not violated; F = the pertinent construct is violated; Mag = Numerical Magnitude; Size = Physical Size.

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Consider two models M_1 and M_2 where M_1 is a special case of M_2. Let L be the likelihood associated with M_i. The null hypothesis for this test is that the special case model is correct. The test statistic χ^2 is defined from the likelihood ratio L_1/L_2 to be:

$$\chi^2 = -2 \ln \left(\frac{L_1}{L_2} \right).$$

This test statistics is χ^2 distributed and its degrees of freedom equal the difference between M_1 and M_2 free parameters. Thus, the general model will be preferred over the restricted model if the test statistic is significant.

Not all the tested models are necessarily nested. In such cases, if the two models have the same number of parameters, then the model with the larger likelihood is chosen. If the two models have different number of parameters, then the Akaike Information Criterion, AIC (Akaike, 1983), adjusted for the number of free parameters, is used. Each model has an associated AIC that is related to its likelihood, L, and a number of parameters, v,

$$\mathrm{AIC} = -2 \ln L + 2v.$$

The results of the hierarchical model fitting procedure are presented in Table 8. As can be noted, the most basic GRT model (i.e., perceptual independence, perceptual separability, and decisional separability) provided the lowest likelihood estimates, whereas more elaborated models containing more free parameters have yielded larger likelihood values. All tested models incorporated the assumption of decisional separability (Silbert & Thomas, 2013).

The dimensions of numerical magnitude and physical size combine to form Stroop-like stimuli. Hence, various models that incorporated violations of perceptual independence in congruent, incongruent, or both types of stimuli, were tested. Interestingly, the best-fit estimates to the data were achieved with models of perceptual dependence within incongruent stimuli. This result agrees well with the dramatic failures of sampling independence observed in the incongruent stimuli. Elaborated models that included violations of perceptual separability showed even better fit. Note, however, that higher likelihood values might reflect greater complexity rather than improved ability of a model to capture real aspects of the data. To address this problem, I employed the principles of model selection described earlier.

To decide between models in a hierarchical relationship I employed the χ^2 procedure (see also, Thomas, 2001a, 2001b). The goal was to find the most general GRT model that yielded a significant improvement in fit over a restricted model in the hierarchy. This model best accounted for the observer's data. Table 9 presents the outcome of these tests. As can be noted, the majority of paired comparisons between general and restricted models in the hierarchy have shown reliable values of the χ^2 statistics, entailing improvement in fit. For example, a model with 10 parameters representing violations of perceptual independence within the two incongruent stimuli, and violations of perceptual separability on the two dimensions, showed superior fit over a model with eight parameters in which perceptual independence was violated on the two incongruent stimuli and perceptual separability was violated on dimension A.

The parameters of the best fitting GRT models for each observer are shown in Table 10. Graphical depictions of these models can be seen in Figure 3. A glimpse at Table 1 confirms the high level of agreement between the observed and predicted response frequencies. Notably, the level of correspondence between the hierarchical model fits and the nonparametric and macro-signal detection tests is remarkable. Inferences made on the basis of former tests were supported by the fits from the hierarchical model fitting procedure. High level of consistency has been also observed in recent GRT studies (Copeland & Wenger, 2006; Cornes et al., 2011; Fitousi & Wenger, 2013; Thomas, 2001a).

In summary, Observer 1, who performed with the numerals 3 and 7, exhibited violations of perceptual independence in the two incongruent stimuli in tandem with violations of perceptual separability on the dimensions of numerical magnitude and physical size. Observer 2, who performed with the numerals 2 and 8, evinced violations of perceptual independence in the two incongruent stimuli and violations of perceptual separability on the dimension of physical size. Observer 3, who was presented with the numerals 2 and 8, exhibited violations of perceptual independence in the two incongruent stimuli and violations of perceptual separability on both dimensions. Finally, Observer 4, who performed with the numerals 3 and 7, exhibited violations of perceptual independence in the incongruent stimuli, and violations of perceptual separability on the dimension of physical size.

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Table 8. Summary of fit statistics for the GRT hierarchical models

		Obs. 1		Obs. 2		Obs. 3		Obs. 4	
Model	Par.	$-\ln L$	AIC	$-\ln L$	AIC	$-\ln L$	AIC	$-\ln L$	AIC
PI, PS, DS	4	1,173.29	2,354.58	1,323.71	2,655.42	1,388.83	2,785.66	1,690.21	3,388.41
PI, $\overline{\text{PS}_a}$, DS	6	1,168.15	2,348.30	1,322.07	2,656.13	1,385.55	2,783.10	1,689.56	3,391.12
PI, $\overline{\text{PS}_b}$, DS	6	1,160.69	2,333.38	1,293.66	2,599.32	1,338.61	2,689.22	1,679.61	3,371.22
PI, $\overline{\text{PS}}$, DS	8	1,155.55	2,327.10	1,271.97	2,559.94	1,361.01	2,738.02	1,678.97	3,373.94
$\overline{\text{PI}}_{1\rho}$, PS, DS	5	1,165.56	2,341.11	1,301.87	2,613.74	1,368.92	2,747.84	1,675.47	3,360.94
$\overline{\text{PI}}_{4\rho}$, PS, DS	8	1,155.33	2,326.66	1,285.00	2,586.00	1,359.58	2,735.16	1,641.45	3,298.90
$\overline{\text{PI}}_{2\text{incong}_\rho}$, PS, DS	6	1,155.78	2,323.55	1,289.74	2,591.48	1,365.49	2,742.98	1,648.05	3,308.10
$\overline{\text{PI}}_{2\text{incong}_\rho}$, $\overline{\text{PS}}$, DS	10	1,137.57	2,295.14	1,257.46	2,534.92	1,335.95	2,691.91	1,635.51	3,291.02
$\overline{\text{PI}}_{2\text{incong}_\rho}$, PS, DS	6	1,172.86	2,361.72	1,318.63	2,649.26	1,383.42	2,778.84	1,683.62	3,379.24
$\overline{\text{PI}}_{2\text{incong}_\rho}$, $\overline{\text{PS}}$, DS	10	1,155.38	2,330.76	1,287.60	2,595.2	1,355.95	2,731.90	1,672.40	3,364.80
$\overline{\text{PI}}_{4\rho}$, $\overline{\text{PS}}$, DS	12	1,140.38	2,300.76	1,277.95	2,575.90	1,350.45	2,722.90	1,639.46	3,298.92
$\overline{\text{PI}}_{2\text{incong}_\rho}$, $\overline{\text{PS}_a}$, DS	8	1,150.09	2,322.18	1,286.97	2,589.94	1,359.53	2,735.06	1,646.10	3,308.20
$\overline{\text{PI}}_{2\text{incong}_\rho}$, $\overline{\text{PS}_b}$, DS	8	1,142.63	2,301.26	1,258.57	2,533.14	1,339.23	2,694.46	1,636.15	3,288.30

Notes. PI = Perceptual Independence; PS = Perceptual Separability; DS = Decisional Separability; Par. = Parameter; AIC = Akaike Information Criterion; $-\ln L$ = $-$Log Likelihood. A bar over a construct implies a violation. Types of violations of perceptual independence are specified by the number of stimuli in which the correlation coefficient rho is a free parameter and the congruency level (cong vs. incong) of those stimuli.

The results from the four observers are highly consistent. Performance with different pairs of numerals has yielded similar GRT models. It is likely that the conclusions drawn are not confined to a particular set of stimuli. In addition, virtually all observers exhibited violations of perceptual independence in incongruent stimuli. This suggests that the numerals were not processed as mere shapes, rather they invoked their semantic (e.g., magnitude) meaning. There is no particular reason to expect the emergence of such congruity effects had the numerals been processed on the basis of their figural characteristics.

The GRT results suggest that the interaction between the dimensions of physical size and numerical magnitude has strong perceptual sources due to: (a) within-stimuli violations of perceptual independence with the incongruent stimuli, and (b) violations of perceptual separability that are more prominent on the physical size dimension than on the numerical magnitude dimension. Taken together, these modeling results entail that physical size and numerical magnitude interact at two early perceptual loci. One occurs within the object, the other at the level of the stimuli ensemble. This set of outcomes confers validation support for the strong version of the analogical hypothesis (Piazza et al., 2007; Pinel et al., 2004).

General Discussion

The current study proposes a novel approach to modeling and testing the perception of numerical information. The approach is based on the application of the GRT (Ashby

Table 9. χ^2 Tests for nested General Recognition Theory Models

		Obs. 1			Obs. 2			Obs. 3			Obs. 4		
General	Restricted	χ^2	*df*	*p*	χ^2	*df*	*p*	χ^2	*df*	*p*	χ^2	*df*	*p*
PI, $\overline{\text{PS}_a}$, DS	PI, PS, DS	10.28	2	.01	3.29	2	.19	6.56	2	.04	1.29	2	.52
PI, $\overline{\text{PS}_b}$, DS	PI, PS, DS	25.20	2	.00	60.10	2	.00	100.44	2	.00	21.18	2	.00
PI, $\overline{\text{PS}}$, DS	PI, $\overline{\text{PS}_a}$, DS	10.28	2	.01	3.29	2	.19	6.56	2	.04	1.29	2	.52
$\overline{\text{PI}}_{1\rho}$, PS, DS	PI, PS, DS	15.47	1	.00	43.68	1	.00	39.82	1	.00	29.47	1	.00
$\overline{\text{PI}}_{4\rho}$, PS, DS	PI, PS, DS	35.92	4	.00	77.42	4	.00	58.50	4	.00	97.51	4	.00
$\overline{\text{PI}}_{2\text{in}_\rho}$, $\overline{\text{PS}}$, DS	PI, PS, DS	35.03	2	.00	67.94	2	.00	46.68	2	.00	84.31	2	.00
$\overline{\text{PI}}_{2\text{in}_\rho}$, $\overline{\text{PS}}$, DS	PI, $\overline{\text{PS}}$, DS	35.96	2	.00	29.02	2	.00	50.12	2	.00	86.91	2	.00
$\overline{\text{PI}}_{2c_\rho}$, $\overline{\text{PS}}$, DS	PI, $\overline{\text{PS}}$, DS	0.34	1	.56	−31.32	1	1.00	10.12	1	.00	13.14	1	.00
$\overline{\text{PI}}_{4_\rho}$, $\overline{\text{PS}}$, DS	$\overline{\text{PI}}_{2c_\rho}$, $\overline{\text{PS}}$, DS	−5.62	2	1	−40.0	1	1.00	−29.00	2	1	−7.89	2	1
$\overline{\text{PI}}_{2\text{in}_\rho}$, $\overline{\text{PS}}$, DS	$\overline{\text{PI}}_{2\text{in}_\rho}$, $\overline{\text{PS}_a}$, DS	25.04	2	.00	59.02	2	.00	47.16	2	.00	21.17	2	.00
$\overline{\text{PI}}_{2\text{in}_\rho}$, $\overline{\text{PS}}$, DS	$\overline{\text{PI}}_{2\text{in}_\rho}$, $\overline{\text{PS}_b}$, DS	10.12	2	.01	2.22	2	.33	6.56	2	.04	1.27	2	.53

Notes. Obs. = Observer; PI = Perceptual Independence; PS = Perceptual Separability; DS = Decisional Separability. A bar over a construct implies a violation. Types of violations of perceptual independence are specified by the number of stimuli in which the correlation coefficient rho is a free parameter and the congruency level (cong vs. incong) of those stimuli.

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Table 10. Parameter estimates for the Best-Fitting Model for each observer

Stimulus (parameter)	Obs. 1	Obs. 2	Obs. 3	Obs. 4
A_1B_1				
ρ	0.00	0.00	0.00	0.00
A_1B_2				
ρ	−0.71	−0.71	−0.51	−0.63
μ_x	−0.05	0.00	0.00	0.00
μ_y	1.78	1.32	1.15	0.67
A_2B_1				
ρ	−0.28	−0.51	−0.53	−0.58
μ_x	2.77	2.71	2.51	1.83
μ_y	−0.22	−0.68	−0.61	−0.48
A_2B_2				
ρ	0.00	0.00	0.00	0.00
μ_x	3.49	2.71	2.83	1.83
μ_y	1.21	0.99	0.89	0.71
γ_x	1.37	1.31	1.30	1.02
γ_y	0.12	0.15	0.34	0.66

& Townsend, 1986) to symbolic and nonsymbolic attributes. All analyses were performed at the level of individual observers. The loci of the cross-talk between the dimensions are at a within-stimulus level (i.e., violations of perceptual independence), and at the level of the stimuli-set (i.e., violations of perceptual separability). Perceptual independence was violated only in incongruent stimuli, thus entailing the activation of semantic values. This latter result points to early perceptual loci of the interaction between symbolic and nonsymbolic attributes, thereby supporting the strong version of the analogical hypothesis (Pinel et al., 2004; Schwarz & Ischebeck, 2003).

According to the strong version of the analogical hypothesis (Piazza et al., 2007) symoblic and nonsymbolic numbers are transformed onto analog magnitude code. This code is shared with other physical attributes such as physical size or luminance. The contribution of the present work is in revealing the internal representations of such attributes and their interactions in the psychological space of the observer. In many respects, the current effort has links to the work of Shepard et al. (1975), who investigated the internal representation of numbers using a multidimensional scaling (MDS) framework. In the MDS approach, the psychological space is derived from similarity ratings of pairs of numerals. Ashby and Perrin (1988) have shown that the GRT and MDS psychological spaces are mathematically comparable, and can be unified under the GRT framework.

One important comment refers to assessment of decisional separability. The potential contribution of decisional mechanisms to the representation of numerical magnitude and physical size has not been estimated in the current work. This is due to the methodological limitations that

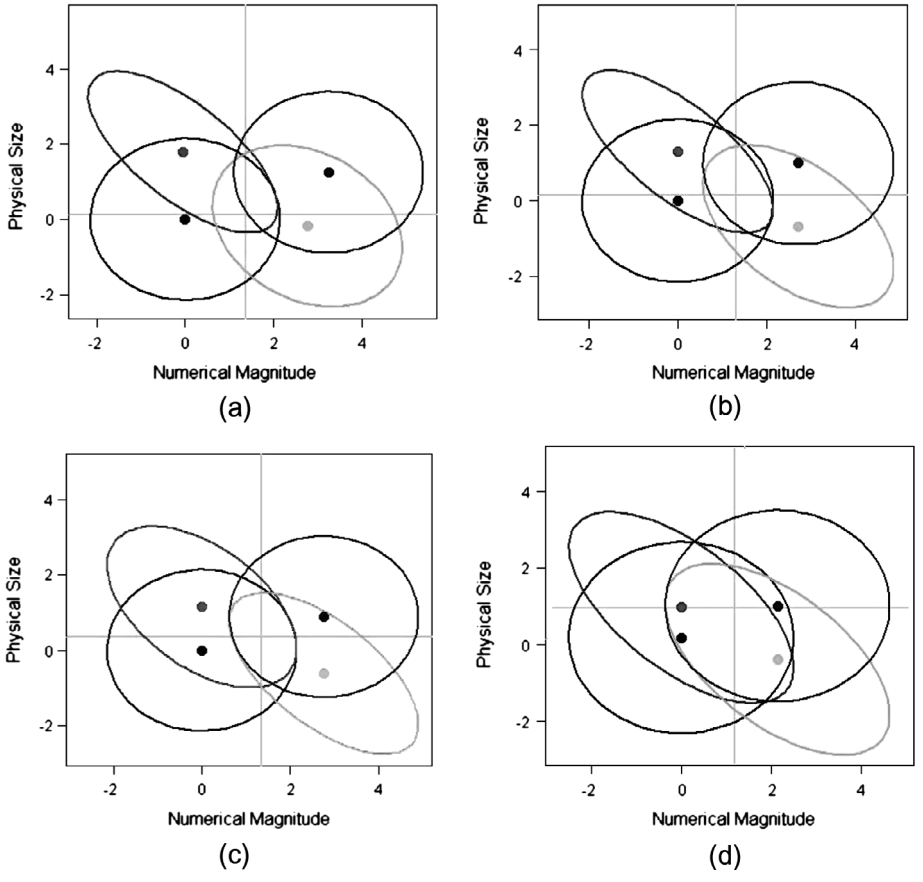


Figure 3. Best-fitting GRT models for the four observers.

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have been discovered recently by various researchers (Mack et al., 2011; Silbert & Thomas, 2013). These authors have shown that decisional separability is not identifiable in Gaussian GRT models. Thus, in the current work, the assumption of decisional separability has been made throughout. It should be noted, though, that while this assumption poses a certain limitation on the number of possible GRT models that can be tested, it improves the power and validity of the available methods. Future work should seek to overcome this limitation.

Dehaene (2007) proposed a general model of numerical cognition that integrates insights from accumulated work in the field (Cordes et al., 2001; Dehaene et al., 1990; Gallistel & Gelman, 1992; Link, 1990; Nieder et al., 2002; Oeffelen & Vos, 1982; Piazza et al., 2007; Shepard et al., 1975; Verguts et al., 2005). Each numerosity/number symbol in the model is represented as a log-normal distribution with the same internal variability for all numbers. Decisional processes are incorporated into this model by adopting a random-walk, and signal detection concepts. In this form, the model is well suited for explaining reaction time data (Link, 1990). The current GRT modeling adopts similar ideas. Each number in the GRT space is assumed to exert a perceptual effect that varies in magnitude from trial-to-trial having a central tendency and variability. This effect is captured by the marginal uni-dimensional normal Gaussian distributions associated with each numerical magnitude. In a similar fashion, each physical size assumes similar variable internal representation. The joint perceptual effects of the two dimensions are captured by bivariate-Gaussian distributions.

The current work may contribute to our understanding of the Stroop phenomenon (Stroop, 1935), one of the most studied phenomena in modern psychology (MacLeod, 1991). In the Stroop task observers are asked to name the ink color of color words while ignoring the semantic meaning of the carrier word. Performance is slower and more error prone with incongruent (the word BLUE is printed in red) than with congruent (the word GREEN is printed in green) stimuli. The application of GRT to numerical magnitude and physical size taps Stroop-like relations among semantic and nonsemantic attributes that carry the quality of congruency. Canonical theories of Stroop have postulated various mechanisms: response conflict (Kahneman, 1973), relative speed of processing (Dyer, 1973; Schwarz & Ischebeck, 2003), automaticity (Logan, 1980), contextual-informational channel interactions (Melara & Algom, 2003), or neuronal interference (Cohen, Dunbar, & McClelland, 1990). These theories often do not model within-stimulus Stroop effects, as indicated in the current study by violations of perceptual independence.

The implications of the present results for the Stroop effect are profound. They show that the differences between congruent and incongruent stimuli are much deeper than the absence or presence of a semantic clash. Incongruent stimuli are processed via the highly interactive architecture, whereas congruent stimuli are processed via separate architecture. This difference in architecture extends to Stroop-like stimuli, notably to those producing the size congruity effect.

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References

Akaike, H. (1983). Information measures and model selection. *Bulletin of the International Statistical Institute, 50*, 277–290.

Algom, D. (1992). Memory psychophysics: An examination of its perceptual and cognitive prospects. In D. Algom (Ed.), *Psychophysical approaches to cognition* (pp. 1–12). Amsterdam, The Netherland: Elsevier.

Algom, D., Dekel, A., & Pansky, A. (1996). The perception of number from the separability of the stimulus: The Stroop effect revisited. *Memory & Cognition, 50*, 557–572.

Ashby, F. G. (1992). Multivariate probability distributions. In F. G. Ashby (Ed.), *Multidimensional models of perception and cognition* (pp. 1–34). Hillsdale, NJ: Erlbaum.

Ashby, F. G., & Lee, W. W. (1991). Predicting similarity and categorization from identification. *Journal of Experimental Psychology: General, 120*, 150–172.

Ashby, F. G., & Maddox, W. T. (1994). A Response Time Theory of separability and integrality in speeded classification. *Journal of Mathematical Psychology, 38*, 423–466.

Ashby, F. G., Maddox, W. T., & Lee, W. W. (1994). On the dangers of averaging across subjects when using multidimensional scaling or the similarity choice model. *Psychological Science, 5*, 144–151.

Ashby, F. G., & Perrin, N. A. (1988). Toward a unified theory of similarity and recognition. *Psychological Review, 95*, 124–150.

Ashby, F. G., & Townsend, J. T. (1986). Varieties of perceptual independence. *Psychological Review, 93*, 154–179.

Ben-Nathan, M. B., Shaki, S., Salti, M., & Algom, D. (2009). Numbers and space: Associations and dissociations. *Psychonomic Bulletin & Review, 16*, 578–582.

Besner, D., & Coltheart, M. (1979). Ideographic and alphabetic processing in skilled reading of English. *Neuropsychologia, 17*, 467–472.

Brannon, E. M., & de Walle, G. A. V. (2001). The development of ordinal numerical competence in young children. *Cognitive Psychology, 43*, 53–81.

Cohen, J. D., Dunbar, K., & McClelland, J. L. (1990). On the control of automatic processes: A parallel distributed processing model of the Stroop effect. *Psychological Review, 117*, 332–361.

Copeland, A. M., & Wenger, M. J. (2006). An investigation of perceptual and decisional influences on the perception of hierarchical forms. *Perception, 35*, 511–529.

Cordes, S., Gelman, R., Gallistel, C. R., & Whalen, J. (2001). Variability signatures distinguish verbal from nonverbal counting for both large and small numbers. *Psychonomic Bulletin & Review, 8*, 698–707.

Cornes, K., Donnelly, N., Godwin, H., & Wenger, M. J. (2011). Perceptual and decisional factors affecting the detection of the Thatcher illusion. *Journal of Experimental Psychology: Human Perception and Performance, 37*, 645–668.

Dehaene, S. (1989). The psychophysics of numerical comparison: A re-examination of apparently incompatible data. *Perception & Psychophysics, 45*, 557–566.

Dehaene, S. (1997). *The number sense: How the mind creates mathematics*. New York, NY: Oxford University Press.

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Dehaene, S. (2007). Symbols and quantities in parietal cortex: Elements of a mathematical theory of number representation and manipulation. In P. Haggard, Y. Rossetti, & M. Kawato (Eds.), *Sensorimotor foundation of higher cognition, volume xxii of attention and performance* (pp. 524–574). Cambridge, MA: Harvard University Press.

Dehaene, S., & Akhavein, R. (1995). Attention, automaticity, and levels of representation in number processing. *Journal of Experimental Psychology: Learning, Memory & Cognition, 21*, 314–326.

Dehaene, S., Bossinin, S., & Giraux, P. (1993). The mental representation of parity and number magnitude. *Journal of Experimental Psychology: General, 122*, 371–396.

Dehaene, S., Dupoux, E., & Mehler, J. (1990). Is numerical comparison digital? Analogical and symbolic effects in two-digit number comparison. *Journal of Experimental Psychology: Human Perception & Performance, 16*, 626–641.

Dyer, F. N. (1973). The Stroop phenomenon and its use in the study of perceptual, cognitive and response processes. *Memory & Cognition, 1*, 106–120.

Estes, W. K. (1956). The problem of inference from curves based on group data. *Psychological Review, 53*, 134–140.

Fias, W., Lammertyn, J., Reynvoet, B., Dupont, P., & Orban, G. (2003). Parietal represenatation of symbolic and nonsymbolic magnitude. *Journal of Cognitive Neuroscience, 15*, 47–56.

Fitousi, D. (2010). Dissociating between cardinal and ordinal and between the value and size magnitudes of coins. *Psychonomic Bulletin & Review, 17*, 889–894.

Fitousi, D. (2013). Mutual information, perceptual independence, and holistic face perception. *Attention, Perception & Psychophysics, 75*, 983–1000.

Fitousi, D., & Algom, D. (2006). Size congruity effects with two-digit numbers: Expanding the number line? *Memory & Cognition, 34*, 445–457.

Fitousi, D., Shaki, S., & Algom, D. (2009). The role of parity, physical size and magnitude in numerical cognition: The SNARC effect revisited. *Attention, Perception & Psychophysics, 71*, 143–155.

Fitousi, D., & Wenger, M. J. (2011). Processing capacity under perceptual and cognitive load: A closer look at load theory. *Journal of Experimental Psychology: Human Perception and Performance, 37*, 781–798.

Fitousi, D., & Wenger, M. J. (2013). Variants of independence in the perception of facial identity and expression. *Journal of Experimental Psychology: Human Perception, and Performance, 39*, 133–155.

Foltz, G. S., Poltrock, S. E., & Potts, G. R. (1984). Mental comparisons of size and magnitude: Size congruity effects. *Journal of Experimental Psychology: Learning, Memory & Cognition, 10*, 442–453.

Gallistel, C. R., & Gelman, R. (1992). Preverbal and verbal counting and computation. *Cognition, 44*, 43–74.

Garner, W. R. (1974). *The processing of information and structure*. New York, NY: Wiley.

Garner, W. R., & Morton, J. (1969). Perceptual independence: Definitions, models, and experimental paradigms. *Psychological Bulletin, 72*, 233–259.

Girelli, L., Lucangeli, D., & Butterworth, B. (2000). The development of automaticity in accessing number magnitude. *Journal of Experimental Child Psychology, 76*, 104–122.

Gourevitch, V., & Galanter, E. (1967). A significance test for one parameter isosensitivity functions. *Psychometrika, 32*, 25–33.

Green, D. M., & Swets, J. A. (1966). *Signal detection theory and psychophysics*. New York, NY: Wiley.

Henik, A., & Tzelgov, J. (1982). Is three greater than five: The relation between physical and semantic size in comparative tasks. *Memory & Cognition, 10*, 389–395.

Johnson, D. M. (1939). Confidence and speed in the two-category judgment. *Archive of Psychology, 241*, 1–52.

Kadlec, H., & Townsend, J. T. (1992a). Implications of marginal and conditional detection parameters for the separabilities and independence of perceptual dimensions. *Journal of Mathematical Psychology, 36*, 325–374.

Kadlec, H., & Townsend, J. T. (1992b). Signal detection analysis of dimensional interactions. In F. G. Ashby (Ed.), *Multidimensional models of perception and cognition* (pp. 181–228). Hillsdale, NJ: Erlbaum.

Kahneman, D. (1973). *Attention and effort*. Englewood Cliffs, NJ: Prentice-Hall.

Link, S. W. (1990). Modelling imageless thought: The relative judgment theory of numerical comparison. *Journal of Mathematical Psychology, 34*, 2–41.

Logan, G. D. (1980). Attention and automaticity in Stroop and priming tasks: Theory and data. *Cognitive Psychology, 12*, 523–553.

Macho, S. (2007). Feature sampling in detection: Implication for the measurement of perceptual independence. *Journal of Experimental Psychology: General, 136*, 133–153.

Mack, M., Richler, J., Gauthier, I., & Palmeri, T. (2011). Indecision on decisional separability. *Psychological Bulletin & Review, 18*, 1–9.

MacLeod, C. M. (1991). Half a century of research on the Stroop effect: An integrative review. *Psychological Bulletin, 109*, 163–203.

Macmillan, N. A., & Creelman, C. D. (1991). *Detection theory: A user's guide*. Cambridge, UK: Cambridge University Press.

Macmillan, N. A., & Creelman, C. D. (2005). *Detection theory: A user's guide* (2nd ed.). Mahwah, NJ: Erlbaum.

Macromedia Authorware (Version 7) [Computer software]. San Francisco, CA: Adobe.

Maddox, W. T. (1992). Perceptual and decisional separability. In F. G. Ashby (Ed.), *Multidimensional models of perception and cognition* (pp. 147–180). Hillsdale, NJ: Erlbaum.

Maddox, W. T. (1999). On the dangers of averaging across observers when comparing decision bound models and generalized context models of categorization. *Perception & Psychophysics, 61*, 354–374.

Maddox, W. T. (2001). Separating perceptual processes from decisional processes in identification and categorization. *Perception & Psychophysics, 63*, 1183–1200.

Maddox, W. T., & Ashby, F. G. (1996). Perceptual separability, decisional separability, and the identification-speeded classification relationship. *Journal of Experimental Psychology: Human Perception and Performance, 22*, 795–817.

Melara, R. D., & Algom, D. (2003). Driven by information: A tectonic theory of Stroop effects. *Psychological Review, 110*, 422–471.

Moyer, R. S. (1973). Comparing objects in memory – evidence suggesting an internal psychophysics. *Perception & Psychophysics, 13*, 180–184.

Moyer, R. S., & Landauer, T. K. (1967). Time required for judgments of numerical inequality. *Nature, 215*, 1519–1520.

Nieder, A., Freedman, D. J., & Miller, E. K. (2002). Representation of the quantity of visual items in the primate prefrontal cortex. *Science, 297*, 1708–1711.

Oeffelen, M. P. V., & Vos, P. G. (1982). A probabilistic model for the discrimination of visual number. *Perception & Psychophysics, 32*, 163–170.

Pansky, A., & Algom, D. (1999). Stroop and Garner effects in comparative judgment of numerals: The role of attention. *Journal of Experimental Psychology: Human Perception and Performance, 25*, 39–59.

Pansky, A., & Algom, D. (2002). Comparative judgment of numerosity and numerical magnitude: Attention preempts automaticity. *Journal of Experimental Psychology: Learning, Memory & Cognition*,(25), 259–274.

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Pavio, A. (1975). Perceptual comparisons through the mind's eys. *Memory & Cognition, 3*, 636–647.
Piazza, M., Pinel, P., Bihan, D. L., & Dehaene, S. (2007). A magnitude code common to numerosities and number symbols in human intraparietal cortex. *Neuron, 53*, 293–305.
Pica, P., Lemer, C., Izard, V., & Dehaene, S. (2004). Exact and approximate arithmetic in an Amazonian indigene group. *Science, 306*, 499–503.
Pinel, P., Piazza, M., Bihan, D. L., & Dehaene, S. (2004). Distributed and overlapping cerebral representations of number, size, and luminance during comparative judgments. *Neuron, 41*, 983–993.
Restle, F. (1970). Speed of adding and comparing numbers. *Experimental Psychology, 83*, 274–278.
Richler, J. J., Gauthier, I., Wenger, M. J., & Palmeri, T. J. (2008). Holistic processing of faces: Perceptual and decisional components. *Journal of Experimental Psychology: Learning, Memory & Cognition, 34*, 328–342.
Schwarz, W., & Heinze, H. J. (1998). On the interaction of numerical and size information in digit comparison: A behavioral and event-related potential study. *Neuropsychologia, 36*, 1167–1179.
Schwarz, W., & Ischebeck, A. (2003). On the relative speed account of number-size interference in comparative judgments of numerals. *Journal of Experimental Psychology: Human, Perception & Performance, 29*, 507–522.
Shepard, R. N., Kilpatrick, D. W., & Cunningham, J. P. (1975). The internal representation of number. *Cognitive Psychology, 1*, 82–138.
Silbert, N. H., & Thomas, R. D. (2013). Decisional separability, model identification, and statistical inference in the general recognition theory framework. *Psychonomic Bulletin & Review, 20*, 1–20.
Stroop, J. R. (1935). Studies of interference in serial verbal reactions. *Journal of Experimental Psychology, 18*, 643–662.
Takahashi, A., & Green, D. W. (1983). Numerical judgments with kanji and kana. *Neuropsychologia, 21*, 259–263.
Tanner, T. A. (1956). Theory of recognition. *Journal of the Acoustical Society of America, 28*, 882–888.
Thomas, R. D. (1995). Gaussian general recognition theory and perceptual independence. *Psychological Review, 102*, 192–200.
Thomas, R. D. (2001a). Characterizing perceptual interactions in face identification using multidimensional signal detection theory. In M. J. Wenger & J. T. Townsend (Eds.), *Computational, geometric, and process perspectives on facial cognition: Contexts and challenges* (pp. 193–228). Mahwah, NJ: Erlbaum.
Thomas, R. D. (2001b). Perceptual interactions of facial dimensions in speeded classification and identification. *Perception & Psychophysics, 63*, 625–650.
Townsend, J. T., Hu, G. G., & Ashby, F. G. (1981). A test of visual feature sampling independence with orthogonal straight lines. *Psychological Research, 43*, 259–275.
Townsend, J. T., Hu, G. G., & Evans, R. J. (1984). Modeling feature perception in brief displays with evidence for positive interdependencies. *Perception & Psychophysics, 36*, 35–49.
Townsend, J. T., Hu, G. G., & Kadlec, H. (1988). Feature sensitivity, bias, and interdependencies as a function of intensity and payoffs. *Perception & Psychophysics, 43*, 575–591.
Townsend, J. T., & Snodgrass, J. G. (1977). Masking effects of similar and dissimilar letters: A test of the interactive channels model. *Perceptual and Motor Skills, 44*, 1212–1228.
Tzelgov, J., Meyer, J., & Henik, A. (1992). Automatic and intentional processing of numerical information. *Journal of Experimental Psychology: Learning, Memory & Cognition, 18*, 166–179.
Vaid, J. (1985). Numerical size comparisons in a phonologically transparent script. *Perception & Psychophysics, 37*, 592–595.
Vaid, J., & Corina, D. (1989). Visual field asymmetries in numerical size comparisons of digits, words, and signs. *Brain and Language, 36*, 117–126.
Verguts, T., & Fias, W. (2004). Representation of number in animals and humans: A neural model. *Journal of Cognitive Neuroscience, 16*, 1493–1504.
Verguts, T., Fias, W., & Stevens, M. (2005). A model of exact small-number represenation. *Psychonomic Bulletin & Review, 12*, 66–80.
Wenger, M. J., & Ingvalson, E. M. (2002). A decisional component of holistic encoding. *Journal of Experimental Psychology: Learning, Memory & Cognition, 28*, 872–892.
Wenger, M. J., & Ingvalson, E. M. (2003). Preserving informational separability and violating decisional separability in facial perception and recognition. *Journal of Experimental Psychology: Learning, Memory & Cognition, 29*, 1106–1118.
Wynn, K. (1992a). Addition and subtraction by human infants. *Nature, 358*, 749–750.
Wynn, K. (1992b). Children's acquisition of the number words and the counting system. *Cognitive Psychology, 24*, 220–251.

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